First ask whether the data agree with thermodynamics

Optional background page 1 | Not compulsory pre-reading. Use selected paragraphs within the 180-minute lesson if needed.

Start the required lesson at week03-study.html. The slides, practice record and exact lab guide provide the full route. No extra reading session is required.

From a fitted curve to a thermodynamic constraint

In Week 2, VLE observations were used to infer γ_i and fit a model. The ideal-vapor relation $\gamma_i = y_i P/(x_i P_i^{sat})$ also assumes negligible Poynting correction and valid pure-component property data. Small fitting errors show reproduction under those assumptions. Thermodynamic consistency asks a separate question: are the inferred properties mutually compatible with thermodynamic identities?

At fixed T and P, the liquid coefficients obey the Gibbs-Duhem constraint:

$$x_1 \, d \ln \gamma_1 + (1-x_1) d \ln \gamma_2 = 0$$

Both coefficients are derivatives of one excess Gibbs-energy function. They cannot be adjusted independently at every composition. For a general path, the right-hand side becomes $-h^E \, dT/(RT^2) + v^E \, dP/(RT)$, using molar excess enthalpy and volume. Isothermal VLE is not necessarily isobaric. The present teaching analysis neglects the excess-volume pressure contribution.

Global and local questions reveal different defects

The area approach integrates $L = \ln(\gamma_1/\gamma_2)$ over the full composition range. Contributions of opposite sign can cancel. A local derivative diagnostic instead compares the derivative of $g^E/(RT)$ with L, with a temperature correction where needed. Numerical derivatives are sensitive to spacing and noise, and an average of local magnitudes may dilute a narrow defect. A plotted residual can therefore be informative even when a summary threshold is met.

Keep four questions separate

Question	Evidence
Did the solver converge?	Numerical stopping and residual checks
Do thermodynamic constraints agree?	Applicable consistency diagnostics and assumptions
Does the model reproduce the data?	Fitting residuals for the selected model
Does it predict new observations?	Evaluation on data excluded from fitting

A consistency test is not a vote among badges. State the dataset, method, numerical evidence, applicable path and limitations. A missing correction or endpoint interval cannot be repaired by renaming “not assessed” as “passed.”

A worked area example and a local counterexample

Optional background page 2 | Synthetic endpoint limits are supplied, not inferred from sparse experimental data.

Integrate five known values

x_1	0	0.25	0.50	0.75	1
L	2	1	0	-1	-2

The trapezoidal contributions are 0.375, 0.125, -0.125 and -0.375. Their sum is S=0. Positive area and negative-area magnitude are each 0.5. Adding 0.20 to every L gives S=0.20, because the composition interval has unit length. The signed S differs from normalized D=100|A-B|/(A+B).

An error can have zero total area

Let f(x)=0.25 cos(2πx). Add (1-x)f to ln γ_1 and -xf to ln γ_2. The weighted sum x ln γ_1+(1-x)ln γ_2 is unchanged, but L increases by f. Its full-range added area is zero. Nevertheless, the derivative of that weighted sum no longer matches L locally. This is the mechanism behind the canceling-error case in Lab 03.

A localized Gaussian defect is a second lesson. It can produce a visible peak while both the area criterion and averaged local score remain below their teaching thresholds. Inspect the shape and its location as well as the score. The controlled defects are intentionally nonphysical constructions for testing diagnostics, not measurements of a real mixture.

Other methods need their own interpretation

Herington: an empirical boiling-range allowance for isobaric data, using J=150(Tmax-Tmin)/Tmin in kelvin and the D-J criterion. This allowance is not independently measured h^E. It does not apply to the required isothermal exercise.

Van Ness-type modeling: compare observed pressure and vapor composition with a selected consistent activity model. The Lab 03 variant fits NRTL, with α=0.47 fixed. It asks whether that model reproduces the data. A failure may reflect the model or property approximation; a small residual does not establish prediction.

Local differential diagnostic: the lab uses three-point finite differences, not NIST TDE's fitted Padé point-test implementation. On a real isobaric path without independent h^E, the uncorrected residual has no thermodynamic verdict.

Reference descriptions: NIST area / Herington, NIST point test, NIST Van Ness. The lab states its departures from those implementations in “Methods” and “Sources & scope.”

Then ask which liquid state is stable

Optional background page 3 | Consistency comes before stability, but neither question replaces the other.

A consistent model may predict two liquids

At fixed T and P, stable equilibrium minimizes total Gibbs energy over admitted states. Equal chemical potentials identify candidate coexisting compositions. Solver convergence, equal slopes or a local minimum alone does not establish the global minimum. A Gibbs-Duhem-consistent model can have a nonconvex mixing-Gibbs curve.

The common tangent compares one liquid with a split

For two coexisting compositions x^{α} and x^{β} on a molar Gibbs-energy curve g(x), the common slope equals $[g(x^{\beta})-g(x^{\alpha})]/(x^{\beta}-x^{\alpha})$. The connecting tangent must support the entire admitted curve from below. A line between the endpoints gives the Gibbs energy attainable by dividing material between those two liquids.

Negative curvature rules out local stability of a homogeneous liquid. Positive curvature does not guarantee global stability: a homogeneous state may be metastable because a finite split still lowers its Gibbs energy. The labels “unstable” and “metastable” describe the hypothetical homogeneous feed, not the final stable two-liquid state.

A material balance determines phase amounts

$f\beta=(z_1-x^{\alpha})/(x^{\beta}-x^{\alpha})$, $f\alpha=1-f\beta$; $z_1=f\alpha x^{\alpha}+f\beta x^{\beta}$

For illustrative supported endpoints 0.20 and 0.80, $z_1=0.50$ gives equal fractions. Changing z_1 to 0.35 gives $f\beta=0.25$ and $f\alpha=0.75$. Check $0.75(0.20)+0.25(0.80)=0.35$. The endpoints stay fixed at the same coexistence conditions; the amount ratio changes.

Lab 04 uses a different numerical model: symmetric Margules A=3. Its calculated endpoints are about 0.070720 and 0.929280, so $z_1=0.35$ gives $f\beta\approx0.325289$. Do not copy the illustrative hand endpoints into this model. Report which values were supplied and which were computed.

Use this bridge in the required lesson

Finish Lab 03 first. After the break, complete practice F by hand, then follow lab-guide page 3 for Lab 04. Use the actual computed endpoints and independently close the material balance. Finish with H1-H4, the Session 3 exit ticket and feedback. LLE fitting and VLLE remain optional extensions.

Original course explanation and examples. Textbook reading locator from the supplied contents: sections 11.1-11.2, 14.11 and 16.2-16.4 for the stability bridge. Method descriptions for consistency are linked on page 2. No textbook text or figures are reproduced.